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地理信息 影像与格网数据内容模型存储规则

Geographic information—
Storage rules of content model for imagery and gridded data

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前 言

本文件按照 GB/T 1.1—2020《标准化工作导则 第 1 部分：标准化文件的结构和起草规则》的规定起草。

本文件与 GB/T 35653.1—2017《地理信息 影像与格网数据的内容模型及编码规则 第 1 部分：内容模型》共同支撑影像与格网数据生产、储存和管理。

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本文件由中华人民共和国自然资源部提出。

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本文件起草单位：武汉大学、深圳大学、北京吉威时代软件股份有限公司、国家基础地理信息中心、武大吉奥信息技术有限公司、北京超图软件股份有限公司、天津市测绘院。

本文件主要起草人：龚健雅、王艳东、高文秀、何友、黄俊韬、张元杰、贺楷锴、李文龙、葛亮。



引 言

本文件采用 ISO/TS 19139-1:2019《地理信息 XML 模式实现 第 1 部分:编码规则》中定义的编码规则将 GB/T 35653.1—2017 中定义的内容模型 UML 映射为 XML 模式,基于通用的数据存储格式定义该内容模型的存储规则,便于影像与格网数据的生产、存储和管理。

地理信息 影像与格网数据内容模型存储规则

1 范围

本文件规定了影像与格网数据内容模型的存储方法与要求、内容编码规则以及 XML 模式实现。
本文件适用于影像与格网数据的生产、存储和管理。



2 规范性引用文件

下列文件中的内容通过文中的规范性引用而构成本文件必不可少的条款。其中，注日期的引用文件，仅该日期对应的版本适用于本文件；不注日期的引用文件，其最新版本（包括所有的修改单）适用于本文件。

GB/T 35653.1—2017 地理信息 影像与格网数据的内容模型及编码规则 第1部分：内容模型

3 术语和定义

GB/T 35653.1—2017 界定的术语和定义适用于本文件。

4 缩略语

下列缩略语适用于本文件。

GeoTIFF：嵌入地理参考信息的标签图像文件格式 (Geographic Tagged Image File Format)

HDF5：分层数据格式 (Hierarchical Data Format Release 5)

TIFF：标签图像文件格式 (Tagged Image File Format)

UML：统一建模语言 (Unified Modeling Language)

XML：可扩展标记语言 (Extensible Markup Language)

XS：XML 模式 (XML Schema)

XSD：XML 模式定义 (XML Schema Definition)

5 存储方法与要求

5.1 存储方法

影像与格网数据的存储方法是将 UML 内容模型转换为 XML 模式，依据 XML 模式实例化后形成 XML 文件，标准的影像数据文件应由数据体文件和 XML 文件共同组成，见图 1。

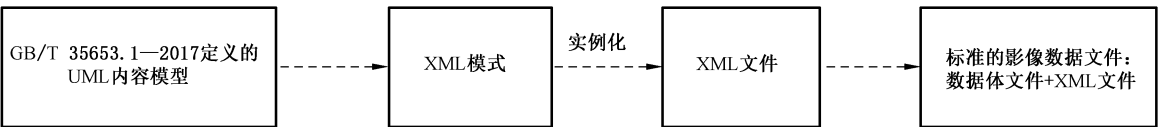


图 1 影像与格网数据的存储模式

5.2 存储要求

基于现有的影像和格网数据格式,包括 TIFF 格式、联合图像专家组 2000(JPEG2000)格式、HDF5 格式等,创建关联的 XML 文件,用于存储 GB/T 35653.1—2017 中定义的内容模型,辅助说明数据体的相关内容。

该 XML 文件的命名规则为:数据文件名.35653.xml。最终标准的数据文件应由数据体文件和该 XML 文件组成。

注:影像与格网数据的存储方法一致(GB/T 35653.1—2017 统一了两者的内容模型)。

示例:具有完整空间位置信息的 3 种常规格式影像与格网数据存储的标准化扩展示例如下。

- a) TIFF/GeoTIFF 格式影像:sample.tif;扩展后:sample.tif,sample.35653.xml。
- b) JPEG2000 格式影像:sample.jp2;扩展后:sample.jp2,sample.35653.xml。
- c) HDF5 格式影像:sample.h5;扩展后:sample.h5,sample.35653.xml。

6 内容编码规则

6.1 总体要求

UML 内容模型映射为 XML 模式包括 3 种映射编码:默认编码、全局要素编码和属性类型编码。按照这 3 种编码形成的影像与格网数据内容模型的 XML 模式应符合附录 A 的规定,影像与格网数据内容模型 XML 描述实例见附录 B。

6.2 默认编码



默认编码是 UML 类映射为 XML 类的基本编码,包括对 UML 类的属性、关联、聚合、组合等关系的编码。规则如下。

- a) 将 UML 类转化为 XML 类。以 xs:complexType 标签作为类型标识,该标签具有 name 属性,在 XML 类默认编码方式中,该属性命名规则为“类名_Type”;如果是抽象类,应增加“Abstract”前缀。

示例 1:以图 2 为例,IE_ImageryandGriddedData 是一个抽象 UML 类

<xs:complexType name="AbstractIE_ImageryandGriddedData_Type" abstract="true">

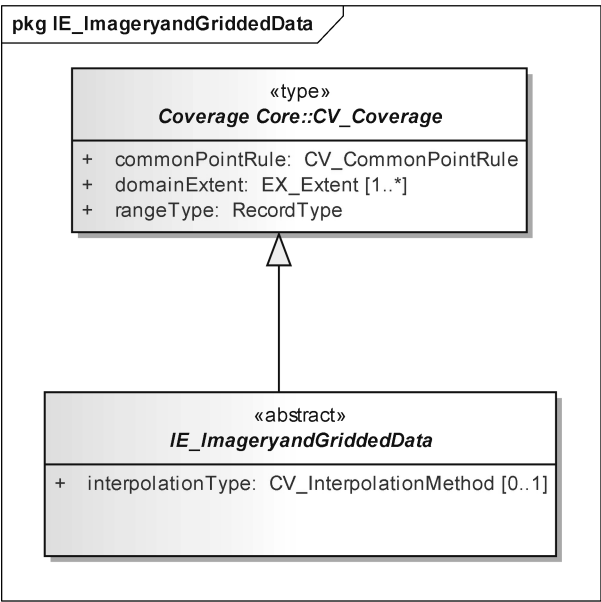


图 2 IE_ImageryandGriddedData UML 模型局部(GB/T 35653.1—2017,8.2.1)

b) 添加子标签 `xs:complexContent` ,用以标记定义 UML 类属性内容的起始标签。

示例 2:

```
<xs:complexType name="AbstractIE_ImageryandGriddedData_Type" abstract="true">
  <xs:complexContent>
```

c) 添加子标签 `xs:extension`,通过其 `base` 属性定义 UML 类的基类;若该类没有基类,`base` 的默认属性值为 `gco:AbstractObject_Type`。

示例 3:IE_ImageryandGriddedData 类的基类是 CV_Coverage

```
<xs:complexType name="AbstractIE_ImageryandGriddedData_Type" abstract="true">
  <xs:complexContent>
    <xs:extension base="gcv:CV_Coverage_Type">
```

d) 添加子标签 `xs:sequence`,用于定义 UML 类的所有属性。该标签下包含多个子标签 `xs:element`,分别定义 UML 类的各个属性。每个 `xs:element` 标签包含 `name`(属性名)属性、`type`(类型)属性等,属性值的形式为“类型_PropertyType”。`xs:element` 标签还包含属性 `maxOccurs`、`minOccurs`,用来约束 UML 类的属性最多和必须出现的次数。

示例 4:interpolationType 是 IE_ImageryandGriddedData 类的一个属性,记录其所用内插方法的类型,该属性最多只能出现 1 次,也可以不出现

```
<xs:complexType name="AbstractIE_ImageryandGriddedData_Type" abstract="true">
  <xs:complexContent>
    <xs:extension base="gcv:CV_Coverage_Type">
      <xs:sequence>
        <xs:element name="interpolationType" type="gcv:CV_InterpolationMethod_PropertyType"
minOccurs="0" maxOccurs="1"/>
```

e) 为前述 a)~d)步添加的标签添加关闭标签。

示例 5:图 2 的完整 XML 编码

```
<xs:complexType name="AbstractIE_ImageryandGriddedData_Type" abstract="true">
  <xs:complexContent>
    <xs:extension base="gcv:CV_Coverage_Type">
      <xs:sequence>
        <xs:element name="interpolationType" type="gcv:CV_InterpolationMethod_PropertyType"
minOccurs="0" maxOccurs="1"/>
      </xs:sequence>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>
```

6.3 全局要素编码

作为全局要素的 UML 类,在将其转换为 XML 类时,应遵循全局要素编码。全局要素编码规则如下。

应编码为包含 `name` 和 `type` 属性的 XS 元素(`xs:element`),`name` 属性值与 UML 类名相同,`type` 属性值为“类名_Type”。

示例:

```
<xs:element name="IE_OpticalImage" type="gie:IE_OpticalImage_Type"/>
```

6.4 属性类型编码

当 UML 类作为其他类的属性时,应使用属性类型编码。规则如下。

- a) 以 `xs:complexType` 标签开始,与 XML 类默认编码的区别在于其 `name` 属性值命名规则为“类名_PropertyType”。

示例 1:以图 3 为例,IE_ImageryandGriddedData 通过 georeferenceableSpatialInfo 角色包含由 IE_Georeferenceable 类定义的经过地理几何校正的影像的相关信息。

```
<xs:complexType name="IE_Georeferenceable_PropertyType">
```

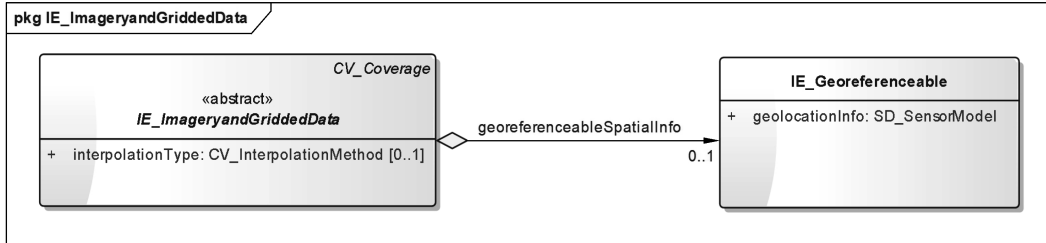


图 3 IE_ImageryandGriddedData UML 模型局部(GB/T 35653.1—2017,8.2.1)

- b) 添加子标签 `xs:sequence`,通过包含一个 `xs:element` 标签定义该类的具体信息,其包含 `ref` 与 `minOccurs` 属性,`ref` 属性值指向已经定义的该 UML 类,`minOccurs` 约束该类定义的属性必须出现的次数,等于 0 时说明该属性是可选的。

示例 2:

```
<xs:complexType name="IE_Georeferenceable_PropertyType">
  <xs:sequence>
    <xs:element ref="gie:IE_Georeferenceable" minOccurs="0"/>
  </xs:sequence>
```

- c) 添加子标签 `xs:attributeGroup`,用于定义 UML 类的各个属性。该属性值为 `gco:ObjectReference`。如果该类定义的属性是可选的,添加 `xs:attribute` 标签,包含 `ref` 属性,其属性值为 `gco:nilReason`,以说明该属性可以接受空值。

示例 3:

```
<xs:complexType name="IE_Georeferenceable_PropertyType">
  <xs:sequence>
    <xs:element ref="gie:IE_Georeferenceable" minOccurs="0"/>
  </xs:sequence>
  <xs:attributeGroup ref="gco:ObjectReference"/>
  <xs:attribute ref="gco:nilReason"/>
```

- d) 添加关闭标签。

示例 4:图 3 的完整 XML 编码

```
<xs:complexType name="IE_Georeferenceable_PropertyType">
  <xs:sequence>
    <xs:element ref="gie:IE_Georeferenceable" minOccurs="0"/>
  </xs:sequence>
  <xs:attributeGroup ref="gco:ObjectReference"/>
  <xs:attribute ref="gco:nilReason"/>
</xs:complexType>
```


附 录 A
(规范性)

影像与格网数据内容模型的 XML 模式

A.1 命名空间

表 A.1 规定了 GB/T 35653.1—2017 中定义的 UML 模型转换为 XML 模式时需要用到的相关命名空间。第一列给出命名空间的简称,第二列是命名空间的中文描述,第三列是命名空间实际的 URI,最后一列标明命名空间的来源。其中 gie 是本文件为 GB/T 35653.1—2017 规定的命名空间。

表 A.1 XML 模式相关命名空间

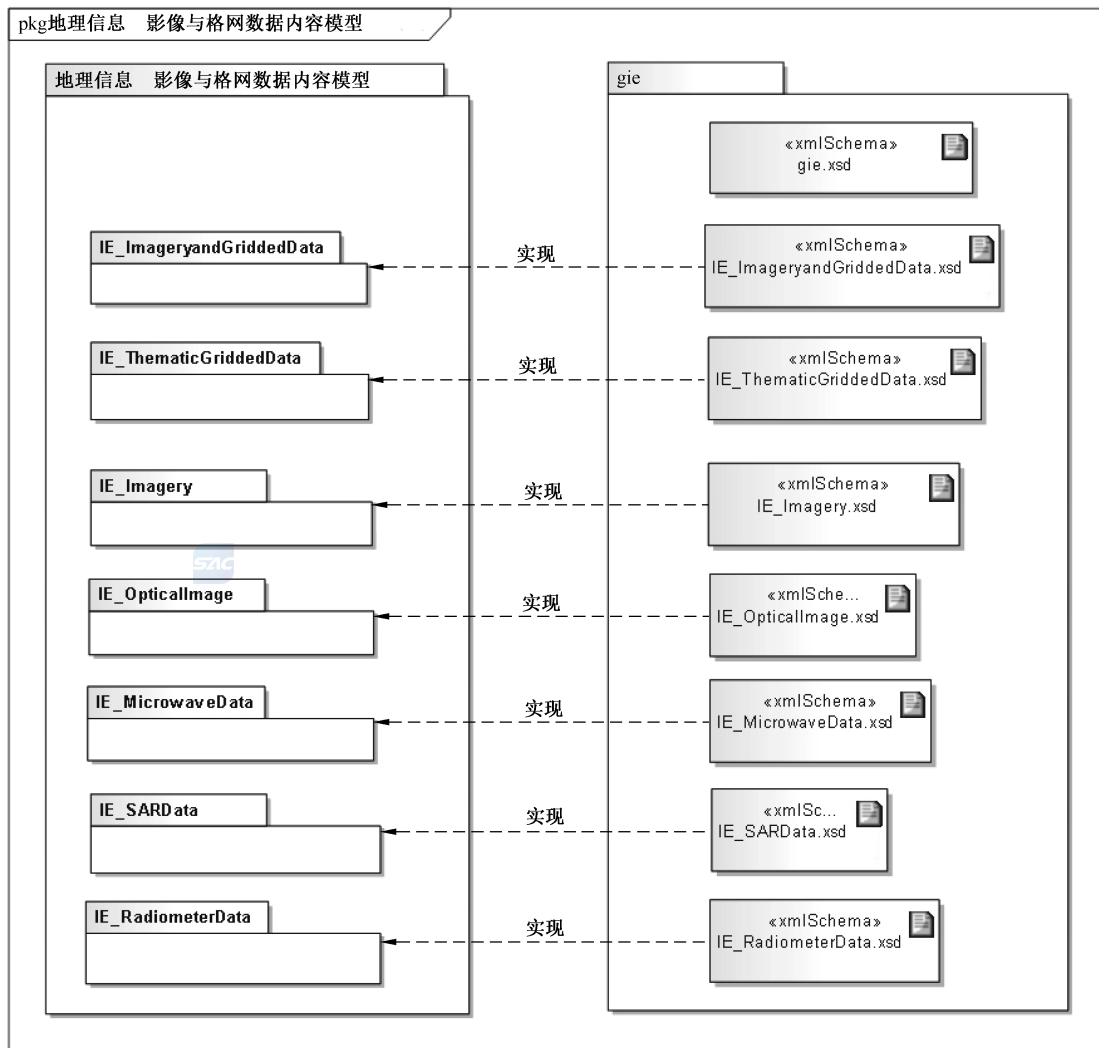
简称	中文描述	命名空间	来源
gie	影像与格网数据内容模型	http://www.lmars.edu.cn/2017/gie * ¹⁾	GB/T 35653.1—2017
gco	概念模式语言	https://schemas.isotc211.org/19103/-/gco/1.2	GB/T 35647—2017
msr	元数据—空间表征	https://schemas.isotc211.org/19115/-1/msr/1.3	ISO 19115-1:2014
mrc	元数据—内容信息	https://schemas.isotc211.org/19115/-1/mrc/1.3	ISO 19115-1:2014
gex	元数据—范围	<code>xmlns:gex="https://schemas.isotc211.org/19115/-1/gex/1.3"</code>	ISO 19115-1:2014
gml	地理标记语言(GML)	http://www.opengis.net/gml/3.2	GB/T 23708—2009
gcv	覆盖的几何与函数模式	http://www.isotc211.org/2005/gcv	ISO 19123:2005
smi	影像传感器的地理定位模型	https://schemas.isotc211.org/19130/-3/smi/1.1	ISO/TS 19130-3:2022
gts	时间模式	http://www.isotc211.org/2005/gts	GB/T 22022—2008
gss	空间模式	http://www.isotc211.org/2005/gss	GB/T 23707—2009
cal	遥感影像传感器的定标与验证	https://schemas.isotc211.org/19159/-1/cal/1.0	ISO/TS 19159-1:2014

A.2 XSD 组织关系

图 A.1 规定了 GB/T 35653.1—2017 中 UML 类与 XML 模式定义文件之间的组织关系。



1) * 为本文件自定义的命名空间。



注：图 A.1 中各 XML 模式定义文件的基本内容如下。

- gie.xsd: 该 xsd 直接实现 gie 的命名空间概念, 不包含声明具体的类型或元素。
- IE_ImageryandGriddedData.xsd: 规定 IE_ImageryandGriddedData, IE_Georectified(继承自 MD_Georectified), IE_Georeferenceable(继承自 MD_Georeferenceable) 的 XML 模式。
- IE_ThematicGriddedData.xsd: 规定 IE_ThematicGriddedData, IE_CategoricalGriddedData, IE_NumericalGriddedData, IE_CategoricalValueAndColour 的 XML 模式。
- IE_Imagery.xsd: 规定 IE_Imagery, IE_SyntheticImage, IE_SimulatedImage, IE_FusedImage, IE_MosaicElement 的 XML 模式。
- IE_OpticalImage.xsd: 规定 IE_OpticalImage 的 XML 模式。
- IE_MicrowaveData.xsd: 规定 IE_MicrowaveData, IE_ActiveMWDData, IE_PassiveMWDData 的 XML 模式。
- IE_SARData.xsd: 规定 IE_SARData, IE_SARComplexData, IE_SARAmplitudeData, IE_SARSensor 的 XML 模式。
- IE_RadiometerData.xsd: 规定 IE_RadiometerData, IE_PMWBand, IE_PMWChannelInfo 的 XML 模式。

图 A.1 UML 类和 XML 模式文件的对应关系

A.3 gie.xsd 文件

gie.xsd 实现了 gie 的命名空间概念, 通过包含的方式集成了各项定义具体类型和元素的 XSD 文

件,对应 GB/T 35653.1—2017 中影像与格网数据内容模型的完整 XML 模式定义。

```
<? xml version = "1.0" encoding = "utf-8"?>
<xs:schema xmlns:xs = "http://www.w3.org/2001/XMLSchema"
xmlns:xlink = "http://www.w3.org/1999/xlink" xmlns:gie = "http://www.lmars.edu.cn/gie"
targetNamespace = "http://www.lmars.edu.cn/gie" elementFormDefault = "qualified"
version = "1.0">
  <xs:include schemaLocation = "../IE_ImageryandGriddedData.xsd"/>
  <xs:include schemaLocation = "../IE_Imagery.xsd"/>
  <xs:include schemaLocation = "../IE_ThematicGriddedData.xsd"/>
  <xs:include schemaLocation = "../IE_MicrowaveData.xsd"/>
  <xs:include schemaLocation = "../IE_OpticalImage.xsd"/>
  <xs:include schemaLocation = "../IE_RadiometerData.xsd"/>
  <xs:include schemaLocation = "../IE_SARData.xsd"/>
</xs:schema>
```

A.4 IE_ImageryandGriddedData.xsd 文件

IE_ImageryandGriddedData.xsd 规定了 GB/T 35653.1—2017 中 UML 类 IE_ImageryandGridded-Data, IE_Georectified 及 IE_Georeferenceable 对应的 XSD 类型和元素。

```
<? xml version = "1.0" encoding = "UTF-8"?>
<xs:schema xmlns:xs = "http://www.w3.org/2001/XMLSchema"
xmlns:gie = "http://www.lmars.edu.cn/2017/gie"
xmlns:msr = "https://schemas.isotc211.org/19115/-1/msr/1.3"
xmlns:gco = "https://schemas.isotc211.org/19103/-/gco/1.2"
xmlns:gcv = "http://www.isotc211.org/2005/gcv"
xmlns:smi = "https://schemas.isotc211.org/19130/-3/smi/1.1"
targetNamespace = "http://www.lmars.edu.cn/2017/gie" elementFormDefault = "qualified"
version = "1.0">
  <xs:import namespace = "https://schemas.isotc211.org/19103/-/gco/1.2"
schemaLocation = "../gco/1.2.0/gco.xsd"/>
  <xs:import namespace = "https://schemas.isotc211.org/19115/-1/msr/1.3"
schemaLocation = "../msr/1.3.0/msr.xsd"/>
  <xs:import namespace = "http://www.isotc211.org/2005/gcv" schemaLocation = "../gcv/gcv.xsd"/>
  <xs:import namespace = "https://schemas.isotc211.org/19130/-3/smi/1.1"
schemaLocation = "../smi/1.1.0/smi.xsd"/>
  <xs:element name = "IE_GeolocationSourceCode" type = "gco:CodeListValue_Type"
substitutionGroup = "gco:CharacterString"/>
  <xs:complexType name = "IE_GeolocationSourceCode_PropertyType">
    <xs:sequence>
      <xs:element ref = "gie:IE_GeolocationSourceCode" minOccurs = "0"/>
    </xs:sequence>
    <xs:attribute ref = "gco:nilReason"/>
  </xs:complexType>
  <xs:complexType name = "IE_Georectified_Type">
    <xs:complexContent>
```

```

    <xs:extension base = "msr:MD_Georectified_Type">
      <xs:sequence>
        <xs:element name = "geolocationSource"
type = "gie:IE_GeolocationSourceCode_PropertyType" minOccurs = "1" maxOccurs = "unbounded"/>
        <xs:element name = "correctionModel"
type = "gie:IE_GeometricCorrectionModelCode_PropertyType"/>
        <xs:element name = "checkPoints" type = "gie:IE_LocationGCP_PropertyType"
minOccurs = "1" maxOccurs = "unbounded"/>
      </xs:sequence>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>
<xs:element name = "IE_Georectified" type = "gie:IE_Georectified_Type"
substitutionGroup = "msr:MD_Georectified"/>
<xs:complexType name = "IE_Georectified_PropertyType">
  <xs:sequence>
    <xs:element ref = "gie:IE_Georectified" minOccurs = "0"/>
  </xs:sequence>
  <xs:attributeGroup ref = "gco:ObjectReference"/>
  <xs:attribute ref = "gco:nilReason"/>
</xs:complexType>
<xs:complexType name = "IE_LocationGCP_Type">
  <xs:choice>
    <xs:element name = "GCPRepository" type = "smi:SD_GCPRepository_PropertyType"/>
    <xs:element name = "imageIdentifiableGCP"
type = "smi:SD_ImageIdentifiableGCP_PropertyType"/>
    <xs:element name = "locationGCP" type = "smi:SD_GCPLocation_PropertyType"/>
  </xs:choice>
</xs:complexType>
<xs:element name = "IE_LocationGCP" type = "gie:IE_LocationGCP_Type"/>
<xs:complexType name = "IE_LocationGCP_PropertyType">
  <xs:sequence>
    <xs:element ref = "gie:IE_LocationGCP" minOccurs = "0"/>
  </xs:sequence>
  <xs:attribute ref = "gco:nilReason"/>
</xs:complexType>
<xs:complexType name = "IE_Georeferenceable_Type">
  <xs:complexContent>
    <xs:extension base = "msr:MD_Georeferenceable_Type">
      <xs:sequence>
        <xs:element name = "geolocationInfo"
type = "smi:SD_SensorModel_PropertyType"/>
      </xs:sequence>
    </xs:extension>
  </xs:complexContent>

```

```

        </xs:extension>
    </xs:complexContent>
</xs:complexType>
<xs:element name = "IE_Georeferenceable" type = "gie:IE_Georeferenceable_Type"
substitutionGroup = "msr:MD_Georeferenceable"/>
<xs:complexType name = "IE_Georeferenceable_PropertyType">
    <xs:sequence>
        <xs:element ref = "gie:IE_Georeferenceable" minOccurs = "0"/>
    </xs:sequence>
    <xs:attributeGroup ref = "gco:ObjectReference"/>
    <xs:attribute ref = "gco:nilReason"/>
</xs:complexType>
<xs:element name = "IE_CompressionSchemaCode" type = "gco:CodeListValue_Type"
substitutionGroup = "gco:CharacterString"/>
<xs:complexType name = "IE_CompressionSchemaCode_PropertyType">
    <xs:sequence>
        <xs:element ref = "gie:IE_GeolocationSourceCode" minOccurs = "0"/>
    </xs:sequence>
    <xs:attribute ref = "gco:nilReason"/>
</xs:complexType>
<xs:element name = "IE_GeometricCorrectionModelCode" type = "gco:CodeListValue_Type"
substitutionGroup = "gco:CharacterString"/>
<xs:complexType name = "IE_GeometricCorrectionModelCode_PropertyType">
    <xs:sequence>
        <xs:element ref = "gie:IE_GeometricCorrectionModelCode" minOccurs = "0"/>
    </xs:sequence>
    <xs:attribute ref = "gco:nilReason"/>
</xs:complexType>
<xs:element name = "IE_LayerSequenceCode" type = "gco:CodeListValue_Type"
substitutionGroup = "gco:CharacterString"/>
<xs:complexType name = "IE_LayerSequenceCode_PropertyType">
    <xs:sequence>
        <xs:element ref = "gie:IE_LayerSequenceCode" minOccurs = "0"/>
    </xs:sequence>
    <xs:attribute ref = "gco:nilReason"/>
</xs:complexType>
<xs:complexType name = "AbstractIE_ImageryandGriddedData_Type" abstract = "true">
    <xs:complexContent>
        <xs:extension base = "gcv:CV_Coverage_Type">
            <xs:sequence>
                <xs:element name = "interpolationType"
type = "gcv:CV_InterpolationMethod_PropertyType" minOccurs = "0" maxOccurs = "1"/>
                <xs:element name = "georectifiedSpatialInfo"

```

```

type = "gie:IE_Georectified_PropertyType" minOccurs = "0" maxOccurs = "1"/>
    <xs:element name = "georeferenceableSpatialInfo"
type = "gie:IE_Georeferenceable_PropertyType" minOccurs = "0" maxOccurs = "1"/>
    </xs:sequence>
    </xs:extension>
    </xs:complexContent>
    </xs:complexType>
    <xs:element name = "AbstractIE_ImageryandGriddedData"
type = "gie:AbstractIE_ImageryandGriddedData_Type" substitutionGroup = "gcv:CV_Coverage"
abstract = "true"/>
    <xs:complexType name = "IE_ImageryandGriddedData_PropertyType">
        <xs:sequence>
            <xs:element ref = "gie:AbstractIE_ImageryandGriddedData" minOccurs = "0"/>
        </xs:sequence>
        <xs:attributeGroup ref = "gco:ObjectReference"/>
        <xs:attribute ref = "gco:nilReason"/>
    </xs:complexType>
    <!-- 以下内容来自于 msr.xsd,用于体现空间分辨率的记录形式 -->
    <!--
    <complexType name = "MD_Georeferenceable_Type">
        <complexContent>
            <extension base = "msr:MD_GridSpatialRepresentation_Type">
                <sequence>
                    <element name = "controlPointAvailability" type = "gco:Boolean_PropertyType"
id = "ISO19115-1.162_ISO19103.Boolean"/>
                    <element name = "orientationParameterAvailability"
type = "gco:Boolean_PropertyType" id = "ISO19115-1.163_ISO19103.Boolean"/>
                    <element name = "orientationParameterDescription"
type = "gco:CharacterString_PropertyType" minOccurs = "0"
id = "ISO19115-1.164_ISO19103.CharacterString"/>
                    <element name = "georeferencedParameters" type = "gco:Record_PropertyType"
id = "ISO19115-1.165_ISO19103.Record"/>
                    <element name = "parameterCitation"
type = "mcc:Abstract_Citation_PropertyType" maxOccurs = "unbounded" minOccurs = "0"
id = "ISO19115-1.166_ISO19115-1.361"/>
                </sequence>
            </extension>
        </complexContent>
    </complexType>

```

```

    <element name = "MD_Georeferenceable" type = "msr:MD_Georeferenceable_Type"
substitutionGroup = "msr:MD_GridSpatialRepresentation" id = "ISO19115-1.161"/>
    <complexType name = "MD_GridSpatialRepresentation_PropertyType">
        <sequence minOccurs = "0">
            <element ref = "msr:MD_GridSpatialRepresentation"/>
        </sequence>
        <attributeGroup ref = "gco:ObjectReference"/>
        <attribute ref = "gco:nilReason"/>
    </complexType>
    <complexType name = "MD_Georectified_Type">
        <complexContent>
            <extension base = "msr:MD_GridSpatialRepresentation_Type">
                <sequence>
                    <element name = "checkPointAvailability" type = "gco:Boolean_PropertyType"
id = "ISO19115-1.154_ISO19103.Boolean"/>
                    <element name = "checkPointDescription"
type = "gco:CharacterString_PropertyType" minOccurs = "0"
id = "ISO19115-1.155_ISO19103.CharacterString"/>
                    <element name = "cornerPoints" type = "gml:PointPropertyType" maxOccurs = "4"
minOccurs = "2" id = "ISO19115-1.156_GM_Point"/>
                    <element name = "centrePoint" type = "gml:PointPropertyType" minOccurs = "0"
id = "ISO19115-1.157_GM_Point"/>
                    <element name = "pointInPixel"
type = "msr:MD_PixelOrientationCode_PropertyType"
id = "ISO19115-1.158_ISO19115-1.MD_PixelOrientationCode"/>
                    <element name = "transformationDimensionDescription"
type = "gco:CharacterString_PropertyType" minOccurs = "0"
id = "ISO19115-1.159_ISO19103.CharacterString"/>
                    <element name = "transformationDimensionMapping"
type = "gco:CharacterString_PropertyType" maxOccurs = "2" minOccurs = "0"
id = "ISO19115-1.160_ISO19103.CharacterString"/>
                </sequence>
            </extension>
        </complexContent>
    </complexType>
    <element name = "MD_Georectified" type = "msr:MD_Georectified_Type"
substitutionGroup = "msr:MD_GridSpatialRepresentation" id = "ISO19115-1.153"/>
    <complexType name = "MD_Georectified_PropertyType">
        <sequence minOccurs = "0">
            <element ref = "msr:MD_Georectified"/>
        </sequence>
        <attributeGroup ref = "gco:ObjectReference"/>
        <attribute ref = "gco:nilReason"/>

```

```

</complexType>
<complexType name = "MD_GridSpatialRepresentation_Type">
  <complexContent>
    <extension base = "msr:AbstractMD_SpatialRepresentation_Type">
      <sequence>
        <element name = "numberOfDimensions" type = "gco:Integer_PropertyType"
id = "ISO19115-1.149_19103.Integer"/>
        <element name = "axisDimensionProperties"
type = "msr:MD_Dimension_PropertyType" maxOccurs = "unbounded" minOccurs = "1"
id = "ISO19115-1.150_19115-1.170"/>
        <element name = "cellGeometry" type = "msr:MD_CellGeometryCode_PropertyType"
id = "ISO19115-1.151_19115-1.MD_CellGeometryCode"/>
        <element name = "transformationParameterAvailability"
type = "gco:Boolean_PropertyType" id = "ISO19115-1.152_19103.Boolean"/>
      </sequence>
    </extension>
  </complexContent>
</complexType>
<element name = "MD_GridSpatialRepresentation"
type = "msr:MD_GridSpatialRepresentation_Type"
substitutionGroup = "msr:AbstractMD_SpatialRepresentation" id = "ISO19115-1.148"/>
<complexType name = "MD_GridSpatialRepresentation_PropertyType">
  <sequence minOccurs = "0">
    <element ref = "msr:MD_GridSpatialRepresentation"/>
  </sequence>
  <attributeGroup ref = "gco:ObjectReference"/>
  <attribute ref = "gco:nilReason"/>
</complexType>
<complexType name = "MD_Dimension_Type">
  <complexContent>
    <extension base = "gco:AbstractObject_Type">
      <sequence>
        <element name = "dimensionName"
type = "msr:MD_DimensionNameTypeCode_PropertyType"
id = "ISO19115-1.171_ISO19115-1.MD_DimensionNameTypeCode"/>
        <element name = "dimensionSize" type = "gco:Integer_PropertyType"
id = "ISO19115-1.172_ISO19103.Integer"/>
        <element name = "resolution" type = "gco:Measure_PropertyType" minOccurs = "0"
id = "ISO19115-1.173_ISO19103.Measure"/>
        <element name = "dimensionTitle" type = "gco:CharacterString_PropertyType"
minOccurs = "0" id = "ISO19115-1.174_ISO19103.CharacterString"/>
        <element name = "dimensionDescription"
type = "gco:CharacterString_PropertyType" minOccurs = "0"

```



```

id = "ISO19115-1.175_ISO19103.CharacterString"/>
    </sequence>
  </extension>
</complexContent>
</complexType>
<element name = "MD_Dimension" type = "msr:MD_Dimension_Type"
substitutionGroup = "gco:AbstractObject" id = "ISO19115-1.170"/>
  <complexType name = "MD_Dimension_PropertyType">
    <sequence minOccurs = "0">
      <element ref = "msr:MD_Dimension"/>
    </sequence>
    <attributeGroup ref = "gco:ObjectReference"/>
    <attribute ref = "gco:nilReason"/>
  </complexType>
-->
</xs:schema>

```

A.5 IE_ThematicGriddedData.xsd 文件

IE_ThematicGriddedData.xsd 规定了 GB/T 35653.1—2017 中 UML 类 IE_ThematicGriddedData, IE_CategoricalGriddedData, IE_NumericalGriddedData 及 IE_CategoricalValueAndColour 对应的 XSD 类型和元素。

```

<? xml version = "1.0" encoding = "UTF-8"?>
<xs:schema xmlns:xs = "http://www.w3.org/2001/XMLSchema"
xmlns:gco = "https://schemas.isotc211.org/19103/-/gco/1.2"
xmlns:mrc = "https://schemas.isotc211.org/19115/-1/mrc/1.3"
xmlns:mrl = "https://schemas.isotc211.org/19115/-1/mrl/1.3"
xmlns:gss = "http://www.isotc211.org/2005/gss"
targetNamespace = "http://www.lmars.edu.cn/2017/gie" elementFormDefault = "qualified"
version = "1.0">
  <xs:import namespace = "https://schemas.isotc211.org/19103/-/gco/1.2"
schemaLocation = "../gco/1.2.0/gco.xsd"/>
  <xs:import namespace = "https://schemas.isotc211.org/19115/-1/mrc/1.3"
schemaLocation = "../mrc/1.3.0/mrc.xsd"/>
  <xs:import namespace = "https://schemas.isotc211.org/19115/-1/mrl/1.3"
schemaLocation = "../mrl/1.3.0/mrl.xsd"/>
  <xs:import namespace = "http://www.isotc211.org/2005/gss"
schemaLocation = "../gss/1.0.0/gss.xsd"/>
  <xs:include schemaLocation = "../IE_ImageryandGriddedData.xsd"/>
  <xs:complexType name = "IE_CategoricalGriddedData_Type">
    <xs:complexContent>
      <xs:extension base = "gie:IE_ThematicGriddedData_Type">
        <xs:sequence>
          <xs:element name = "classificationDescription"

```

```

type = "gco:CharacterString_PropertyType" minOccurs = "0" maxOccurs = "1"/>
    <xs:element name = "numberOfCategories" type = "gco:Integer_PropertyType"
minOccurs = "1" maxOccurs = "1"/>
    <xs:element name = "bitsPerValue" type = "gco:Integer_PropertyType"
minOccurs = "0" maxOccurs = "1"/>
    <xs:element name = "lookupTable"
type = "gie:IE_CategoricalValueAndColour_PropertyType" minOccurs = "0" maxOccurs = "unbounded"/>
    </xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:element name = "IE_CategoricalGriddedData" type = "gie:IE_CategoricalGriddedData_Type"
substitutionGroup = "gie:IE_ThematicGriddedData"/>
<xs:complexType name = "IE_CategoricalGriddedData_PropertyType">
    <xs:sequence>
        <xs:element ref = "gie:IE_CategoricalGriddedData" minOccurs = "0"/>
    </xs:sequence>
    <xs:attributeGroup ref = "gco:ObjectReference"/>
    <xs:attribute ref = "gco:nilReason"/>
</xs:complexType>
<xs:complexType name = "IE_NumericalGriddedData_Type">
    <xs:complexContent>
        <xs:extension base = "gie:IE_ThematicGriddedData_Type">
            <xs:sequence>
                <xs:element name = "valueDescription"
type = "gco:CharacterString_PropertyType" minOccurs = "1" maxOccurs = "1"/>
            </xs:sequence>
        </xs:extension>
    </xs:complexContent>
</xs:complexType>
<xs:element name = "IE_NumericalGriddedData" type = "gie:IE_NumericalGriddedData_Type"
substitutionGroup = "gie:IE_ThematicGriddedData"/>
<xs:complexType name = "IE_NumericalGriddedData_PropertyType">
    <xs:sequence>
        <xs:element ref = "gie:IE_NumericalGriddedData" minOccurs = "0"/>
    </xs:sequence>
    <xs:attributeGroup ref = "gco:ObjectReference"/>
    <xs:attribute ref = "gco:nilReason"/>
</xs:complexType>
<xs:complexType name = "IE_ThematicGriddedData_Type">
    <xs:complexContent>
        <xs:extension base = "gie:AbstractIE_ImageryandGriddedData_type">
            <xs:sequence>

```

```

        <xs:element name = "dataInfo"
type = "mrc:MD_CoverageDescription_PropertyType" minOccurs = "1" maxOccurs = "1"/>
        <xs:element name = "annotation" type = "gss:GM_Object_PropertyType"
minOccurs = "0" maxOccurs = "unbounded"/>
        <xs:element name = "geographicFeature" type = "gss:GM_Object_PropertyType"
minOccurs = "0" maxOccurs = "unbounded"/>
        <xs:element name = "retrievalProcessingInfo"
type = "mrl:LE_Processing_PropertyType" minOccurs = "0" maxOccurs = "unbounded"/>
        <xs:element name = "sourceInfo" type = "gco:CharacterString_PropertyType"
minOccurs = "0" maxOccurs = "unbounded"/>
    </xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:element name = "IE_ThematicGriddedData" type = "gie:IE_ThematicGriddedData_Type"
substitutionGroup = "gie:AbstractIE_ImageryandGriddedData_type"/>
<xs:complexType name = "IE_ThematicGriddedData_PropertyType">
    <xs:sequence>
        <xs:element ref = "gie:IE_ThematicGriddedData" minOccurs = "0"/>
    </xs:sequence>
    <xs:attributeGroup ref = "gco:ObjectReference"/>
    <xs:attribute ref = "gco:nilReason"/>
</xs:complexType>
<xs:complexType name = "IE_ColourPalette_Type">
    <xs:complexContent>
        <xs:extension base = "gco:AbstractObject_Type">
            <xs:sequence>
                <xs:element name = "red" type = "gco:Integer_PropertyType" minOccurs = "1"
maxOccurs = "1"/>
                <xs:element name = "green" type = "gco:Integer_PropertyType" minOccurs = "1"
maxOccurs = "1"/>
                <xs:element name = "blue" type = "gco:Integer_PropertyType" minOccurs = "1"
maxOccurs = "1"/>
            </xs:sequence>
        </xs:extension>
    </xs:complexContent>
</xs:complexType>
<xs:element name = "IE_ColourPalette" type = "gie:IE_ColourPalette_Type"/>
<xs:complexType name = "IE_ColourPalette_PropertyType">
    <xs:sequence>
        <xs:element ref = "gie:IE_ColourPalette" minOccurs = "0"/>
    </xs:sequence>
    <xs:attributeGroup ref = "gco:ObjectReference"/>

```

```

        <xs:attribute ref = "gco:nilReason"/>
    </xs:complexType>
    <xs:complexType name = "IE_CategoricalValueAndColour_Type">
        <xs:complexContent>
            <xs:extension base = "gco:AbstractObject_Type">
                <xs:sequence>
                    <xs:element name = "categoryName"
type = "gco:CharacterString_PropertyType" minOccurs = "1" maxOccurs = "1"/>
                    <xs:element name = "colourPalette"
type = "gie:IE_ColourPalette_PropertyType" minOccurs = "0" maxOccurs = "1"/>
                    <xs:element name = "description" type = "gco:CharacterString_PropertyType"
minOccurs = "0" maxOccurs = "1"/>
                    <xs:element name = "value" type = "gco:Integer_PropertyType" minOccurs = "1"
maxOccurs = "1"/>
                </xs:sequence>
            </xs:extension>
        </xs:complexContent>
    </xs:complexType>
    <xs:element name = "IE_CategoricalValueAndColour"
type = "gie:IE_CategoricalValueAndColour_Type"/>
    <xs:complexType name = "IE_CategoricalValueAndColour_PropertyType">
        <xs:sequence>
            <xs:element ref = "gie:IE_CategoricalValueAndColour" minOccurs = "0"/>
        </xs:sequence>
        <xs:attributeGroup ref = "gco:ObjectReference"/>
        <xs:attribute ref = "gco:nilReason"/>
    </xs:complexType>
</xs:schema>

```

A.6 IE_Imagery.xsd 文件

IE_Imagery.xsd 规定了 GB/T 35653.1—2017 中 UML 类 IE_Imagery, IE_SyntheticImage, IE_SimulatedImage, IE_FusedImage 及 IE_MosaicElement 对应的 XSD 类型和元素。

```

<? xml version = "1.0" encoding = "UTF-8"?>
<xs:schema xmlns:xs = "http://www.w3.org/2001/XMLSchema"
xmlns:gie = "http://www.lmars.edu.cn/2017/gie"
xmlns:mrc = "https://schemas.isotc211.org/19115/-1/mrc/1.3"
xmlns:gex = "https://schemas.isotc211.org/19115/-1/gex/1.3"
xmlns:mri = "https://schemas.isotc211.org/19115/-1/mri/1.3"
xmlns:mac = "https://schemas.isotc211.org/19115/-2/mac/2.2"
xmlns:mrl = "https://schemas.isotc211.org/19115/-1/mrl/1.3"
xmlns:gco = "https://schemas.isotc211.org/19103/-/gco/1.2"
xmlns:cal = "https://schemas.isotc211.org/19159/-1/cal/1.0"
targetNamespace = "http://www.lmars.edu.cn/2017/gie" elementFormDefault = "qualified"

```

```

version = "1.0">
  <xs:import namespace = "https://schemas.isotc211.org/19103/-/gco/1.2"
schemaLocation = "../gco/1.2.0/gco.xsd"/>
  <xs:import namespace = "https://schemas.isotc211.org/19115/-1/mrc/1.3"
schemaLocation = "../mrc/1.3.0/content.xsd"/>
  <xs:import namespace = "https://schemas.isotc211.org/19115/-1/gex/1.3"
schemaLocation = "../gex/1.3.0/extent.xsd"/>
  <xs:import namespace = "https://schemas.isotc211.org/19115/-1/mri/1.3"
schemaLocation = "../mri/1.3.0/identification.xsd"/>
  <xs:import namespace = "https://schemas.isotc211.org/19115/-2/mac/2.2"
schemaLocation = "../mac/2.2.0/acquisitionInformationImagery.xsd"/>
  <xs:import namespace = "https://schemas.isotc211.org/19115/-1/mrl/1.3"
schemaLocation = "../mrl/1.3.0/lineage.xsd"/>
  <xs:import namespace = "https://schemas.isotc211.org/19115/-1/mrl/1.3"
schemaLocation = "../mrl/1.3.0/lineageImagery.xsd"/>
  <xs:import namespace = "https://schemas.isotc211.org/19159/-1/cal/1.0"
schemaLocation = "../cal/1.0.0/cal.xsd"/>
  <xs:include schemaLocation = "../IE_ImageryandGriddedData.xsd"/>
  <xs:complexType name = "IE_MosaicElement_Type">
    <xs:complexContent>
      <xs:extension base = "gco:AbstractObject_Type">
        <xs:sequence>
          <xs:element name = "elementID" type = "gco:Integer_PropertyType"
minOccurs = "1" maxOccurs = "1"/>
          <xs:element name = "acquisitionTime" type = "gts:TM_Period_PropertyType"
minOccurs = "1" maxOccurs = "1"/>
          <xs:element name = "geometry" type = "gex:EX_Extent_PropertyType"
minOccurs = "1" maxOccurs = "1"/>
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:element name = "IE_MosaicElement" type = "gie:IE_MosaicElement_Type"/>
  <xs:complexType name = "IE_MosaicElement_PropertyType">
    <xs:sequence>
      <xs:element ref = "gie:IE_MosaicElement" minOccurs = "0"/>
    </xs:sequence>
    <xs:attributeGroup ref = "gco:ObjectReference"/>
    <xs:attribute ref = "gco:nilReason"/>
  </xs:complexType>
  <xs:element name = "IE_RadiometricCalibrationTypeCode" type = "gco:CodeListValue_Type"
substitutionGroup = "gco:CharacterString"/>
  <xs:complexType name = "IE_RadiometricCalibrationTypeCode_PropertyType">

```

```

    <xs:sequence>
      <xs:element ref = "gie:IE_RadiometricCalibrationTypeCode" minOccurs = "0"/>
    </xs:sequence>
    <xs:attribute ref = "gco:nilReason"/>
  </xs:complexType>
  <xs:complexType name = "IE_SyntheticImage_Type">
    <xs:complexContent>
      <xs:extension base = "gie:AbstractIE_Imagery_Type">
        <xs:sequence>
          <xs:element name = "intendedUse" type = "mri:MD_Usage_PropertyType"
minOccurs = "0" maxOccurs = "1"/>
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:element name = "IE_SyntheticImage" type = "gie:IE_SyntheticImage_Type"
substitutionGroup = "gie:AbstractIE_Imagery"/>
  <xs:complexType name = "IE_SyntheticImage_PropertyType">
    <xs:sequence>
      <xs:element ref = "gie:IE_SyntheticImage" minOccurs = "0"/>
    </xs:sequence>
    <xs:attributeGroup ref = "gco:ObjectReference"/>
    <xs:attribute ref = "gco:nilReason"/>
  </xs:complexType>
  <xs:complexType name = "IE_SimulatedImage_Type">
    <xs:complexContent>
      <xs:extension base = "gie:IE_SyntheticImage_Type">
        <xs:sequence>
          <xs:element name = "simulateMethod"
type = "gco:CharacterString_PropertyType" minOccurs = "1" maxOccurs = "1"/>
          <xs:element name = "provenance" type = "mri:LI_Lineage_PropertyType"
minOccurs = "0" maxOccurs = "1"/>
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:element name = "IE_SimulatedImage" type = "gie:IE_SimulatedImage_Type"
substitutionGroup = "gie:IE_SyntheticImage"/>
  <xs:complexType name = "IE_SimulatedImage_PropertyType">
    <xs:sequence>
      <xs:element ref = "gie:IE_SimulatedImage" minOccurs = "0"/>
    </xs:sequence>
    <xs:attributeGroup ref = "gco:ObjectReference"/>

```

```

    <xs:attribute ref = "gco:nilReason"/>
  </xs:complexType>
  <xs:complexType name = "IE_FusedImage_Type">
    <xs:complexContent>
      <xs:extension base = "gie:IE_SyntheticImage_Type">
        <xs:sequence>
          <xs:element name = "numberOfSourceImages"
type = "gco:Integer_PropertyType" minOccurs = "1" maxOccurs = "1"/>
          <xs:element name = "processingSteps"
type = "mrl:IE_ProcessStep_PropertyType" minOccurs = "0" maxOccurs = "unbounded"/>
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:element name = "IE_FusedImage" type = "gie:IE_FusedImage_Type"
substitutionGroup = "gie:IE_SyntheticImage"/>
  <xs:complexType name = "IE_FusedImage_PropertyType">
    <xs:sequence>
      <xs:element ref = "gie:IE_FusedImage" minOccurs = "0"/>
    </xs:sequence>
    <xs:attributeGroup ref = "gco:ObjectReference"/>
    <xs:attribute ref = "gco:nilReason"/>
  </xs:complexType>
  <xs:complexType name = "AbstractIE_Imagery_Type" abstract = "true">
    <xs:complexContent>
      <xs:extension base = "gie:AbstractIE_ImageryandGriddedData_Type">
        <xs:sequence>
          <xs:element name = "acquisitionTime" type = "gts:TM_Period_PropertyType"
minOccurs = "1" maxOccurs = "1"/>
          <xs:element name = "imageDescription"
type = "mrc:MD_ImageDescription_PropertyType" minOccurs = "0" maxOccurs = "1"/>
          <xs:element name = "isMosaic" type = "gco:Boolean_PropertyType"
minOccurs = "1" maxOccurs = "1"/>
          <xs:element name = "numberOfBands" type = "gco:Integer_PropertyType"
minOccurs = "1" maxOccurs = "1"/>
          <xs:element name = "processingAction"
type = "gie:IE_ProcessingActionCode_PropertyType" minOccurs = "0" maxOccurs = "1"/>
          <xs:element name = "radiometricCalibrationType"
type = "gie:IE_RadiometricCalibrationTypeCode_PropertyType" minOccurs = "0" maxOccurs = "1"/>
          <xs:element name = "sensorCalibrationValidation"
type = "cal:CA_CalibrationValidation_PropertyType" minOccurs = "0" maxOccurs = "1"/>
          <xs:element name = "platformInfo" type = "mac:MI_Platform_PropertyType"
minOccurs = "0" maxOccurs = "1"/>

```

```

        <xs:element name = "sensorInfo" type = "mac:MI_Instrument_PropertyType"
minOccurs = "0" maxOccurs = "1"/>
        <xs:element name = "bandinformation" type = "mrc:MI_Band_PropertyType"
minOccurs = "0" maxOccurs = "unbounded"/>
        <xs:element name = "masaicInfo" type = "gie:IE_MosaicElement_PropertyType"
minOccurs = "1" maxOccurs = "unbounded"/>
    </xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:element name = "AbstractIE_Imagery" type = "gie:AbstractIE_Imagery_Type"
substitutionGroup = "gie:AbstractIE_ImageryandGriddedData" abstract = "true"/>
<xs:complexType name = "IE_Imagery_PropertyType">
    <xs:sequence>
        <xs:element ref = "gie:AbstractIE_Imagery" minOccurs = "0"/>
    </xs:sequence>
    <xs:attributeGroup ref = "gco:ObjectReference"/>
    <xs:attribute ref = "gco:nilReason"/>
</xs:complexType>
<xs:element name = "IE_ProcessingActionCode" type = "gco:CodeListValue_Type"
substitutionGroup = "gco:CharacterString"/>
<xs:complexType name = "IE_ProcessingActionCode_PropertyType">
    <xs:sequence>
        <xs:element ref = "gie:IE_ProcessingActionCode" minOccurs = "0"/>
    </xs:sequence>
</xs:complexType>
<!-- 以下内容来自于 mrc.xsd,用于体现光谱分辨率的记录形式 -->
<!--
<complexType name = "MI_Band_Type">
    <complexContent>
        <extension base = "mrc:MD_Band_Type">
            <sequence>
                <element name = "bandBoundaryDefinition"
type = "mrc:MI_BandDefinition_PropertyType" minOccurs = "0"/>
                <element name = "nominalSpatialResolution" type = "gco:Real_PropertyType"
minOccurs = "0"/>
                <element name = "transferFunctionType"
type = "mrc:MI_TransferFunctionTypeCode_PropertyType" minOccurs = "0"/>
                 <element name = "transmittedPolarisation"
type = "mrc:MI_PolarisationOrientationCode_PropertyType" minOccurs = "0"/>
                <element name = "detectedPolarisation"
type = "mrc:MI_PolarisationOrientationCode_PropertyType" minOccurs = "0"/>
            </sequence>

```



```

        </extension>
      </complexContent>
    </complexType>
    <element name="MI_Band" type="mrc:MI_Band_Type" substitutionGroup="mrc:MD_Band"/>
    <complexType name="MI_Band_PropertyType">
      <sequence minOccurs="0">
        <element ref="mrc:MI_Band"/>
      </sequence>
      <attributeGroup ref="gco:ObjectReference"/>
      <attribute ref="gco:nilReason"/>
    </complexType>
    <xs:complexType name="MD_Band_Type">
      <xs:complexContent>
        <xs:extension base="mrc:MD_RangeDimension_Type">
          <xs:sequence>
            <xs:element name="bitsPerValue" type="gco:Integer_PropertyType"
minOccurs="0" maxOccurs="1"/>
            <xs:element name="boundMax" type="gco:Real_PropertyType" minOccurs="0"
maxOccurs="1"/> --> <! -- 光谱分辨率:最长波长 -->
            <! -- <xs:element name="boundMin" type="gco:Real_PropertyType"
minOccurs="0" maxOccurs="1"/> --> <! -- 光谱分辨率:最短波长 -->
            <! -- <xs:element name="boundUnits" type="gco:UomLength_PropertyType"
minOccurs="0" maxOccurs="1"/>
            <xs:element name="peakResponse" type="gco:Real_PropertyType"
minOccurs="0" maxOccurs="1"/>
            <xs:element name="toneGradation" type="gco:Integer_PropertyType"
minOccurs="0" maxOccurs="1"/>
          </xs:sequence>
        </xs:extension>
      </xs:complexContent>
    </xs:complexType>
    <xs:element name="MD_Band" type="mrc:MD_Band_Type"
substitutionGroup="mrc:MD_SampleDimension"/>
    <complexType name="MI_Band_PropertyType">
      <sequence minOccurs="0">
        <element ref="mrc:MI_Band"/>
      </sequence>
      <attributeGroup ref="gco:ObjectReference"/>
      <attribute ref="gco:nilReason"/>
    </complexType>
  -->
</xs:schema>

```

A.7 IE_OpticalImage.xsd 文件

IE_OpticalImage.xsd 规定了 GB/T 35653.1—2017 中 UML 类 IE_OpticalImage 对应的 XSD 类型和元素。

```
<? xml version = "1.0" encoding = "UTF-8"?>
<xs:schema xmlns:xs = "http://www.w3.org/2001/XMLSchema"
xmlns:gie = "http://www.lmars.edu.cn/2017/gie"
xmlns:gco = "https://schemas.isotc211.org/19103/-/gco/1.2"
targetNamespace = "http://www.lmars.edu.cn/2017/gie" elementFormDefault = "qualified"
version = "1.0">
  <xs:import namespace = "https://schemas.isotc211.org/19103/-/gco/1.2"
schemaLocation = "../gco/1.2.0/gco.xsd"/>
  <xs:include schemaLocation = "./IE_Imagery.xsd"/>
  <xs:complexType name = "IE_OpticalImage_Type">
    <xs:complexContent>
      <xs:extension base = "gie:AbstractIE_Imagery_Type">
        <xs:sequence>
          <xs:element name = "opticalImageType"
type = "gie:IE_OpticalImageTypeCode_PropertyType" minOccurs = "0" maxOccurs = "1"/>
          <xs:element name = "opticalSensorType"
type = "gie:IE_OpticalSensorTypeCode_PropertyType" minOccurs = "0" maxOccurs = "1"/>
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:element name = "IE_OpticalImage" type = "gie:IE_OpticalImage_Type"
substitutionGroup = "gie:AbstractIE_Imagery"/>
  <xs:complexType name = "IE_OpticalImage_PropertyType">
    <xs:sequence>
      <xs:element ref = "gie:IE_OpticalImage" minOccurs = "0"/>
    </xs:sequence>
    <xs:attributeGroup ref = "gco:ObjectReference"/>
    <xs:attribute ref = "gco:nilReason"/>
  </xs:complexType>
  <xs:element name = "IE_OpticalSensorTypeCode" type = "gco:CodeListValue_Type"
substitutionGroup = "gco:CharacterString"/>
  <xs:complexType name = "IE_OpticalSensorTypeCode_PropertyType">
    <xs:sequence>
      <xs:element ref = "gie:IE_OpticalSensorTypeCode" minOccurs = "0"/>
    </xs:sequence>
  </xs:complexType>
  <xs:element name = "IE_OpticalImageTypeCode" type = "gco:CodeListValue_Type"
substitutionGroup = "gco:CharacterString"/>
```

```

<xs:complexType name = "IE_OpticalImageTypeCode_PropertyType">
  <xs:sequence>
    <xs:element ref = "gie:IE_OpticalImageTypeCode" minOccurs = "0"/>
  </xs:sequence>
</xs:complexType>
</xs:schema>

```

A.8 IE_MicrowaveData.xsd 文件

IE_MicrowaveData.xsd 规定了 GB/T 35653.1—2017 中 UML 类 IE_MicrowaveData, IE_ActiveMWDData 及 IE_PassiveMWDData 对应的 XSD 类型和元素。

```

<? xml version = "1.0" encoding = "UTF-8"?>
<xs:schema xmlns:xs = "http://www.w3.org/2001/XMLSchema"
  targetNamespace = "http://www.lmars.edu.cn/2017/gie" elementFormDefault = "qualified"
  version = "1.0">
  <xs:include schemaLocation = "../IE_Imagery.xsd"/>
  <xs:complexType name = "AbstractIE_MicrowaveData_Type" abstract = "true">
    <xs:complexContent>
      <xs:extension base = "gie:AbstractIE_Imagery">
        <xs:sequence/>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:element name = "AbstractIE_MicrowaveData" type = "gie:AbstractIE_MicrowaveData_Type"
    substitutionGroup = "gie:AbstractIE_Imagery" abstract = "true"/>
  <xs:complexType name = "IE_PassiveMWDData_Type">
    <xs:complexContent>
      <xs:extension base = "gie:AbstractIE_MicrowaveData_Type">
        <xs:sequence/>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:element name = "IE_PassiveMWDData" type = "gie:IE_PassiveMWDData_Type"
    substitutionGroup = "gie:AbstractIE_MicrowaveData"/>
  <xs:complexType name = "IE_ActiveMWDData_Type">
    <xs:complexContent>
      <xs:extension base = "gie:AbstractIE_MicrowaveData_Type">
        <xs:sequence/>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:element name = "IE_ActiveMWDData" type = "gie:IE_ActiveMWDData_Type"
    substitutionGroup = "gie:AbstractIE_MicrowaveData"/>
</xs:schema>

```

A.9 IE_SARData.xsd 文件

IE_SARData.xsd 规定了 GB/T 35653.1—2017 中 UML 类 IE_SARData, IE_SARComplexData, IE_SARAmplitudeData 及 IE_SARSensor 对应的 XSD 类型和元素。

```
<? xml version = "1.0" encoding = "UTF-8"?>
<xs:schema xmlns:xs = "http://www.w3.org/2001/XMLSchema"
xmlns:gie = "http://www.lmars.edu.cn/2017/gie"
targetNamespace = "http://www.lmars.edu.cn/2017/gie" elementFormDefault = "qualified"
version = "1.0">
  <xs:import namespace = "https://schemas.isotc211.org/19103/-/gco/1.2"
schemaLocation = "../gco/1.2.0/gco.xsd"/>
  <xs:include schemaLocation = "../IE_MicrowaveData.xsd"/>
  <xs:element name = "IE_SARImagingModeCode" type = "gco:CodeListValue_Type"
substitutionGroup = "gco:CharacterString"/>
  <xs:complexType name = "IE_SARImagingModeCode_PropertyType">
    <xs:sequence>
      <xs:element ref = "gie:IE_SARImagingModeCode" minOccurs = "0"/>
    </xs:sequence>
    <xs:attribute ref = "gco:nilReason"/>
  </xs:complexType>
  <xs:complexType name = "IE_SARAmplitudeData_Type">
    <xs:complexContent>
      <xs:extension base = "gie:IE_SARData_Type">
        <xs:sequence>
          <xs:element name = "calibrationFactor" type = "gco:Real_PropertyType"
minOccurs = "0" maxOccurs = "1"/>
          <xs:element name = "looksInAzimuth" type = "gco:Integer_PropertyType"
minOccurs = "1" maxOccurs = "1"/>
          <xs:element name = "looksInRange" type = "gco:Integer_PropertyType"
minOccurs = "1" maxOccurs = "1"/>
          <xs:element name = "sarSensor" type = "gie:IE_SARSensor_PropertyType"
minOccurs = "0" maxOccurs = "1"/>
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:element name = "IE_SARAmplitudeData" type = "IE_SARAmplitudeData_Type"
substitutionGroup = "gie:IE_SARData"/>
  <xs:complexType name = "IE_SARAmplitudeData_PropertyType">
    <xs:sequence>
      <xs:element ref = "gie:IE_SARAmplitudeData" minOccurs = "0"/>
    </xs:sequence>
    <xs:attributeGroup ref = "gco:ObjectReference"/>
  </xs:complexType>
```

```

    <xs:attribute ref = "gco:nilReason"/>
  </xs:complexType>
  <xs:complexType name = "IE_SARDopplerCentroidParameter_Type">
    <xs:complexContent>
      <xs:extension base = "gco:AbstractObject_Type">
        <xs:sequence>
          <xs:element name = "polynomial_coefficient_0"
type = "gco:Real_PropertyType" minOccurs = "1" maxOccurs = "1"/>
          <xs:element name = "polynomial_coefficient_1"
type = "gco:Real_PropertyType" minOccurs = "1" maxOccurs = "1"/>
          <xs:element name = "polynomial_coefficient_3"
type = "gco:Real_PropertyType" minOccurs = "1" maxOccurs = "1"/>
          <xs:element name = "reference_point" type = "gco:Real_PropertyType"
minOccurs = "1" maxOccurs = "1"/>
          <xs:element name = "time" type = "gco:DateTime_PropertyType" minOccurs = "1"
maxOccurs = "1"/>
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:element name = "IE_SARDopplerCentroidParameter"
type = "IE_SARDopplerCentroidParameter_Type"/>
  <xs:complexType name = "IE_SARDopplerCentroidParameter_PropertyType">
    <xs:sequence>
      <xs:element ref = "gie:IE_SARDopplerCentroidParameter" minOccurs = "0"/>
    </xs:sequence>
    <xs:attributeGroup ref = "gco:ObjectReference"/>
    <xs:attribute ref = "gco:nilReason"/>
  </xs:complexType>
  <xs:complexType name = "IE_OrbitParameter_Type">
    <xs:complexContent>
      <xs:extension base = "gco:AbstractObject_Type">
        <xs:sequence>
          <xs:element name = "time" type = "gco:DateTime_PropertyType" minOccurs = "1"
maxOccurs = "1"/>
          <xs:element name = "velocityX" type = "gco:Real_PropertyType"
minOccurs = "1" maxOccurs = "1"/>
          <xs:element name = "velocityY" type = "gco:Real_PropertyType"
minOccurs = "1" maxOccurs = "1"/>
          <xs:element name = "velocityZ" type = "gco:Real_PropertyType"
minOccurs = "1" maxOccurs = "1"/>
          <xs:element name = "x" type = "gco:Real_PropertyType" minOccurs = "1"
maxOccurs = "1"/>

```

```

        <xs:element name = "y" type = "gco:Real_PropertyType" minOccurs = "1"
maxOccurs = "1"/>
        <xs:element name = "z" type = "gco:Real_PropertyType" minOccurs = "1"
maxOccurs = "1"/>
    </xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:element name = "IE_OrbitParameter" type = "IE_OrbitParameter_Type"/>
<xs:complexType name = "IE_OrbitParameter_PropertyType">
    <xs:sequence>
        <xs:element ref = "gie:IE_OrbitParameter" minOccurs = "0"/>
    </xs:sequence>
    <xs:attributeGroup ref = "gco:ObjectReference"/>
    <xs:attribute ref = "gco:nilReason"/>
</xs:complexType>
<xs:complexType name = "IE_SARSensor_Type">
    <xs:complexContent>
        <xs:extension base = "gco:AbstractObject_Type">
            <xs:sequence>
                <xs:element name = "azimuthBandwidth" type = "gco:Real_PropertyType"
minOccurs = "1" maxOccurs = "1"/>
                <xs:element name = "dopplerCentroid"
type = "gie:IE_SARDopplerCentroidParameter_PropertyType" minOccurs = "1"
maxOccurs = "unbounded"/>
                <xs:element name = "oneWayTimeToFirstRangePixel"
type = "gco:Time_PropertyType" minOccurs = "1" maxOccurs = "1"/>
                <xs:element name = "orbitParameter"
type = "gie:IE_OrbitParameter_PropertyType" minOccurs = "1" maxOccurs = "unbounded"/>
                <xs:element name = "pulseRepititionFrequency"
type = "gco:Real_PropertyType" minOccurs = "1" maxOccurs = "1"/>
                <xs:element name = "rangeBandwidth" type = "gco:Real_PropertyType"
minOccurs = "1" maxOccurs = "1"/>
                <xs:element name = "rangeSamplingRate" type = "gco:Real_PropertyType"
minOccurs = "1" maxOccurs = "1"/>
                <xs:element name = "timeOfFirstAzimuthLine"
type = "gco:DateTime_PropertyType" minOccurs = "1" maxOccurs = "1"/>
            </xs:sequence>
        </xs:extension>
    </xs:complexContent>
</xs:complexType>
<xs:element name = "IE_SARSensor" type = "IE_SARSensor_Type"/>
<xs:complexType name = "IE_SARSensor_PropertyType">

```

```

    <xs:sequence>
      <xs:element ref = "gie:IE_SARSensor" minOccurs = "0"/>
    </xs:sequence>
    <xs:attributeGroup ref = "gco:ObjectReference"/>
    <xs:attribute ref = "gco:nilReason"/>
  </xs:complexType>
  <xs:element name = "IE_SARComplexLayerTypeCode" type = "gco:CodeListValue_Type"
substitutionGroup = "gco:CharacterString"/>
  <xs:complexType name = "IE_SARComplexLayerTypeCode_PropertyType">
    <xs:sequence>
      <xs:element ref = "gie:IE_SARComplexLayerTypeCode" minOccurs = "0"/>
    </xs:sequence>
    <xs:attribute ref = "gco:nilReason"/>
  </xs:complexType>
  <xs:complexType name = "IE_SARData_Type">
    <xs:complexContent>
      <xs:extension base = "gie:IE_ActiveMWData_Type">
        <xs:sequence>
          <xs:element name = "farRangeIncidenceAngle" type = "gco:Real_PropertyType"
minOccurs = "0" maxOccurs = "1"/>
          <xs:element name = "nearRangeIncidenceAngle"
type = "gco:Real_PropertyType" minOccurs = "0" maxOccurs = "1"/>
          <xs:element name = "sarImagingMode"
type = "gie:IE_SARImagingModeCode_PropertyType" minOccurs = "1" maxOccurs = "1"/>
          <xs:element name = "waveLength" type = "gco:Real_PropertyType"
minOccurs = "1" maxOccurs = "1"/>
          <xs:element name = "sarModeDescription"
type = "gco:CharacterString_PropertyType" minOccurs = "0" maxOccurs = "1"/>
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:element name = "IE_SARData" type = "IE_SARData_Type"
substitutionGroup = "gie:IE_ActiveMWData"/>
  <xs:complexType name = "IE_SARData_PropertyType">
    <xs:sequence>
      <xs:element ref = "gie:IE_SARData" minOccurs = "0"/>
    </xs:sequence>
    <xs:attributeGroup ref = "gco:ObjectReference"/>
    <xs:attribute ref = "gco:nilReason"/>
  </xs:complexType>
  <xs:complexType name = "IE_SARComplexData_Type">
    <xs:complexContent>

```

```

    <xs:extension base = "gie:IE_SARData_Type">
      <xs:sequence>
        <xs:element name = "complexLayerTypes"
type = "gie:IE_SARComplexLayerTypeCode_PropertyType" minOccurs = "1" maxOccurs = "1"/>
        <xs:element name = "lineSpacing" type = "gco:CharacterString_PropertyType"
minOccurs = "1" maxOccurs = "1"/>
        <xs:element name = "pixelSpacing"
type = "gco:CharacterString_PropertyType" minOccurs = "1" maxOccurs = "1"/>
        <xs:element name = "sarSensor" type = "gie:IE_SARSensor_PropertyType"
minOccurs = "1" maxOccurs = "1"/>
      </xs:sequence>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>
<xs:element name = "IE_SARComplexData" type = "IE_SARComplexData_Type"
substitutionGroup = "gie:IE_SARData"/>
<xs:complexType name = "IE_SARComplexData_PropertyType">
  <xs:sequence>
    <xs:element ref = "gie:IE_SARComplexData" minOccurs = "0"/>
  </xs:sequence>
  <xs:attribute ref = "gco:nilReason"/>
</xs:complexType>
</xs:schema>

```

A.10 IE_RadiometerData.xsd 文件

IE_RadiometerData.xsd 规定了 GB/T 35653.1—2017 中 UML 类 IE_RadiometerData, IE_PMW-Band 及 IE_PMWChannelInfo 对应的 XSD 类型和元素。

```

<? xml version = "1.0" encoding = "UTF-8"?>
<xs:schema xmlns:xs = "http://www.w3.org/2001/XMLSchema"
xmlns:gie = "http://www.lmars.edu.cn/2017/gie"
targetNamespace = "http://www.lmars.edu.cn/2017/gie" elementFormDefault = "qualified"
version = "1.0">
  <xs:import namespace = "https://schemas.isotc211.org/19103/-/gco/1.2"
schemaLocation = "../gco/1.2.0/gco.xsd"/>
  <xs:include schemaLocation = "../IE_MicrowaveData.xsd"/>
  <xs:element name = "IE_PMWScanningModeCode" type = "gco:CodeListValue_Type"
substitutionGroup = "gco:CharacterString"/>
  <xs:complexType name = "IE_PMWScanningModeCode_PropertyType">
    <xs:sequence>
      <xs:element ref = "gie:IE_PMWScanningModeCode" minOccurs = "0"/>
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name = "IE_PMWBand_Type">

```



```

    <xs:complexContent>
      <xs:extension base = "gco:AbstractObject_Type">
        <xs:sequence>
          <xs:element name = "bandName" type = "gie:IE_PMWBandNameCode_PropertyType"
minOccurs = "1" maxOccurs = "1"/>
          <xs:element name = "numberOfChannels" type = "gco:Integer_PropertyType"
minOccurs = "1" maxOccurs = "1"/>
          <xs:element name = "channels" type = "gie:IE_PMWChannelInfo_PropertyType"
minOccurs = "1" maxOccurs = "1"/>
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:element name = "IE_PMWBand" type = "gie:IE_PMWBand_Type"/>
  <xs:complexType name = "IE_PMWBand_PropertyType">
    <xs:sequence>
      <xs:element ref = "gie:IE_PMWBand" minOccurs = "0"/>
    </xs:sequence>
    <xs:attributeGroup ref = "gco:ObjectReference"/>
    <xs:attribute ref = "gco:nilReason"/>
  </xs:complexType>
  <xs:element name = "IE_PMWBandNameCode" type = "gco:CodeListValue_Type"
substitutionGroup = "gco:CharacterString"/>
  <xs:complexType name = "IE_PMWBandNameCode_PropertyType">
    <xs:sequence>
      <xs:element ref = "gie:IE_PMWBandNameCode" minOccurs = "0"/>
    </xs:sequence>
    <xs:attribute ref = "gco:nilReason"/>
  </xs:complexType>
  <xs:element name = "IE_PMWApertureTypeCode" type = "gco:CodeListValue_Type"
substitutionGroup = "gco:CharacterString"/>
  <xs:complexType name = "IE_PMWApertureTypeCode_PropertyType">
    <xs:sequence>
      <xs:element ref = "gie:IE_PMWApertureTypeCode" minOccurs = "0"/>
    </xs:sequence>
    <xs:attribute ref = "gco:nilReason"/>
  </xs:complexType>
  <xs:complexType name = "IE_RadiometerData_Type">
    <xs:complexContent>
      <xs:extension base = "gie:IE_PassiveMWData_Type">
        <xs:sequence>
          <xs:element name = "scanningMode"
type = "gie:IE_PMWScanningModeCode_PropertyType" minOccurs = "1" maxOccurs = "1"/>

```

```

        <xs:element name = "PMWBand" type = "gie:IE_PMWBand_PropertyType"
minOccurs = "1" maxOccurs = "1"/>
        <xs:element name = "apertureType"
type = "gie:IE_PMWApertureTypeCode_PropertyType" minOccurs = "1" maxOccurs = "1"/>
    </xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:element name = "IE_RadiometerData" type = "gie:IE_RadiometerData_Type"
substitutionGroup = "gie:IE_PassiveMWDData"/>
<xs:complexType name = "IE_RadiometerData_PropertyType">
    <xs:sequence>
        <xs:element ref = "gie:IE_RadiometerData" minOccurs = "0"/>
    </xs:sequence>
    <xs:attributeGroup ref = "gco:ObjectReference"/>
    <xs:attribute ref = "gco:nilReason"/>
</xs:complexType>
<xs:element name = "IE_ChannelPropertyCode" type = "gco:CodeListValue_Type"
substitutionGroup = "gco:CharacterString"/>
<xs:complexType name = "IE_ChannelPropertyCode_PropertyType">
    <xs:sequence>
        <xs:element ref = "gie:IE_ChannelPropertyCode" minOccurs = "0"/>
    </xs:sequence>
    <xs:attribute ref = "gco:nilReason"/>
</xs:complexType>
<xs:complexType name = "IE_PMWChannelInfo_Type">
    <xs:complexContent>
        <xs:extension base = "gco:AbstractObject_Type">
            <xs:sequence>
                <xs:element name = "channelWidth" type = "gco:Real_PropertyType"
minOccurs = "1" maxOccurs = "1"/>
                <xs:element name = "centreFrequency" type = "gco:Real_PropertyType"
minOccurs = "1" maxOccurs = "1"/>
                <xs:element name = "channelProperty"
type = "gie:IE_ChannelPropertyCode_PropertyType" minOccurs = "1" maxOccurs = "1"/>
            </xs:sequence>
        </xs:extension>
    </xs:complexContent>
</xs:complexType>
<xs:element name = "IE_PMWChannelInfo" type = "gie:IE_PMWChannelInfo_Type"/>
<xs:complexType name = "IE_PMWChannelInfo_PropertyType">
    <xs:sequence>
        <xs:element ref = "gie:IE_PMWChannelInfo" minOccurs = "0"/>

```

```
</xs:sequence>  
<xs:attributeGroup ref = "gco:ObjectReference"/>  
<xs:attribute ref = "gco:nilReason"/>  
</xs:complexType>  
</xs:schema>
```

附 录 B

(资料性)

内容模型 XML 描述实例

B.1 光学影像文件的标准内容模型 XML 描述实例

下面的 XML 内容是一个光学影像文件的标准内容模型实例,描述了该光学影像的空间参考、纠正模型、检查点、获取时间和波段等信息。

```
<? xml version = "1.0" encoding = "UTF-8"?>
<gie:IE_OpticalImage xsi:schemaLocation = "http://www.lmars.edu.cn/2017/gie
IE_OpticalImage.xsd"
xmlns:gie = "http://www.lmars.edu.cn/2017/gie"
xmlns:gco = "https://schemas.isotc211.org/19103/-/gco/1.2"
xmlns:gcv = "http://www.isotc211.org/2005/gcv"
xmlns:msr = "https://schemas.isotc211.org/19115/-1/msr/1.3"
xmlns:smi = "https://schemas.isotc211.org/19130/-3/smi/1.1"
xmlns:mrc = "https://schemas.isotc211.org/19115/-1/mrc/1.3"
xmlns:gex = "https://schemas.isotc211.org/19115/-1/gex/1.3"
xmlns:mri = "https://schemas.isotc211.org/19115/-1/mri/1.3"
xmlns:mac = "https://schemas.isotc211.org/19115/-2/mac/2.2"
xmlns:mrl = "https://schemas.isotc211.org/19115/-1/mrl/1.3"
xmlns:gts = "http://www.isotc211.org/2005/gts"
xmlns:gml = "http://www.isotc211.org/gml/3.2"
xmlns:epsg = "urn:x-ogp:spec:schema-xsd:EPSG:1.0:dataset"
xmlns:xsi = "http://www.w3.org/2001/XMLSchema-instance">
  <!-- CV_Coverage -->
  <gcv:commonPointRule>
    <gcv:CV_CommonPointRule
codeList = "http://www.lmars.edu.cn/gie/codeList/gieCodeLists.xml # CV_CommonPointRule"
codeListValue = "average">average</gcv:CV_CommonPointRule>
  </gcv:commonPointRule>
  <gcv:rangeType>
    <gco:RecordType>
      <gco:fieldType>
        <gco:FieldType>
          <gco:fieldName>
            <gco:CharacterString>intensity</gco:CharacterString>
          </gco:fieldName>
          <gco:fieldType>
            <gco:TypeName>
              <gco:aName>
                <gco:CharacterString>Integer</gco:CharacterString>
```

```

        </gco:aname>
        </gco:TypeName>
        </gco:fieldType>
        </gco:FieldType>
    </gco:fieldType>
</gco:RecordType>
</gcv:rangeType>
<gcv:domainExtent>
    <gex:EX_Extent>
        <gex:geographicElement>
            <gex:EX_GeographicBoundingBox>
                <gex:westBoundLongitude>
                    <gco:Decimal>115.5</gco:Decimal>
                </gex:westBoundLongitude>
                <gex:eastBoundLongitude>
                    <gco:Decimal>115.56</gco:Decimal>
                </gex:eastBoundLongitude>
                <gex:southBoundLatitude>
                    <gco:Decimal>32.2</gco:Decimal>
                </gex:southBoundLatitude>
                <gex:northBoundLatitude>
                    <gco:Decimal>32.26</gco:Decimal>
                </gex:northBoundLatitude>
            </gex:EX_GeographicBoundingBox>
        </gex:geographicElement>
    </gex:EX_Extent>
</gcv:domainExtent>
<gcv:CRS><! -- 空间参考信息 -->
    <gml:GeodeticCRS gml:id = "epsg-crs-4490">
        <gml:identifier
gml:codeSpace = "OGP">urn:ogc:def:crs:EPSG::4490</gml:identifier>
            <gml:name>China Geodetic Coordinate System 2000</gml:name>
            <gml:remarks>Adopted July 2008. Replaces Xian 1980 (CRS code 4610).</gml:remarks>
            <gml:scope>Horizontal component of 3D system.</gml:scope>
            <gml:ellipsoidalCS xlink:href = "urn:ogc:def:cs:EPSG::6422" />
            <gml:geodeticDatum xlink:href = "urn:ogc:def:datum:EPSG::1043" />
        </gml:GeodeticCRS>
    </gcv:CRS>
<! -- IE_ImageryandGriddedData -->
<gie:interpolationType>
    <gcv:CV_InterpolationType
codeList = "http://www.lmars.edu.cn/gie/codeList/gieCodeLists.xml#CV_InterpolationType"
codeListValue = "cubic">cubic</gcv:CV_InterpolationType>

```

```

    </gie:interpolationType>
    <gie:georectifiedSpatialInfo> <! -- 影像纠正信息 -->
      <gie:IE_Georectified>
        <gie:geolocationSource>
          <gie:IE_GeolocationSourceCode
gie:codeList = "http://www.lmars.edu.cn/gie/codeList/gieCodeLists.xml # IE_GeolocationSource-
Code" codeListValue = "GCP">GCP</gie:IE_GeolocationSourceCode>
          </gie:geolocationSource>
          <gie:correctionModel>
            <gie:IE_GeometricCorrectionModelCode
gie:codeList = "http://www.lmars.edu.cn/gie/codeList/gieCodeLists.xml # IE_GeometricCorrec-
tionModelCode"
codeListValue = "CorrespondenceModel">CorrespondenceModel</gie:IE_GeometricCorrectionMod-
elCode>
            </gie:correctionModel>
            <gie:checkPoints> <! -- 检查点 -->
              <gie:IE_LocationGCP>
                <gie:locationGCP>
                  <smi:SD_GCPLocation>
                    <smi:locationGeographicPosition>
                      <gml:Point>
                        <gml:pos>115.53
32.23</gml:pos> 
                      </gml:Point>
                    </smi:locationGeographicPosition>
                    <smi:locationGridPosition>
                      <gcv:CV_GridPoint>
                        <gcv:gridCoord>
                          <gcv:coordValues>3000
3000</gcv:coordValues>
                        </gcv:gridCoord>
                      </gcv:CV_GridPoint>
                    </smi:locationGridPosition>
                  </smi:SD_GCPLocation>
                </gie:locationGCP>
              </gie:IE_LocationGCP>
            </gie:checkPoints>
            <! -- MD_Georectified_Type -->
            <msr:numberOfDimensions>
              <gco:Integer>2</gco:Integer>
            </msr:numberOfDimensions>
            <msr:axisDimensionProperties>
              <msr:MD_Dimension>

```

```

        <msr:dimensionName>
            <msr:MD_DimensionNameTypeCode
codeList = "http://www.lmars.edu.cn/gie/codeList/gieCodeLists.xml#MD_DimensionNameTypeCode"
codeListValue = "column">column</msr:MD_DimensionNameTypeCode>
        </msr:dimensionName>
        <msr:dimensionSize>
            <gco:Integer>6000</gco:Integer>
        </msr:dimensionSize>
        <msr:resolution>
            <gco:Measure
gml:uom = "urn:ogc:def:uom:EPSG::9102">0.00001</gco:Measure>
            </msr:resolution>
        </msr:MD_Dimension>
        <msr:MD_Dimension>
            <msr:dimensionName>
                <msr:MD_DimensionNameTypeCode
codeList = "http://www.lmars.edu.cn/gie/codeList/gieCodeLists.xml#MD_DimensionNameTypeCode"
codeListValue = "row">row</msr:MD_DimensionNameTypeCode>
            </msr:dimensionName>
            <msr:dimensionSize>
                <gco:Integer>6000</gco:Integer>
            </msr:dimensionSize>
            <msr:resolution>
                <gco:Measure
gml:uom = "urn:ogc:def:uom:EPSG::9102">0.00001</gco:Measure>
            </msr:resolution>
        </msr:MD_Dimension>
    </msr:axisDimensionProperties>
    <msr:cellGeometry>
        <msr:MD_CellGeometryCode
codeList = "http://www.lmars.edu.cn/gie/codeList/gieCodeLists.xml#MD_CellGeometryCode"
codeListValue = "area">area</msr:MD_CellGeometryCode>
    </msr:cellGeometry>
    <msr:transformationParameterAvailability>
        <gco:Boolean>false</gco:Boolean>
    </msr:transformationParameterAvailability>
    <msr:checkPointAvailability>
        <gco:Boolean>true</gco:Boolean>
    </msr:checkPointAvailability>
    <msr:pointInPixel>
        <msr:MD_PixelOrientationCode>centre</msr:MD_PixelOrientationCode>
    </msr:pointInPixel>
</gie:IE_Georectified>

```

```

</gie:georectifiedSpatialInfo>
<! -- IE_Imagery -->
<gie:acquisitionTime>
  <gts:TM_Period>
    <gts:begin>
      <gts:TM_Position>
        <gts:date8601>2020-08-01</gts:date8601>
      </gts:TM_Position>
    </gts:begin>
    <gts:end>
      <gts:TM_Position>
        <gts:date8601>2020-08-01</gts:date8601>
      </gts:TM_Position>
    </gts:end>
  </gts:TM_Period>
</gie:acquisitionTime>
<gie:isMosaic>
  <gco:Boolean>>false</gco:Boolean>
</gie:isMosaic>
<gie:numberofBands>
  <gco:Integer>1</gco:Integer>
</gie:numberofBands>
<! -- 波段描述 -->
<gie:bandinformation>
  <mrc:MI_Band>
    <mrc:bitsPerValue>
      <gco:Integer>16</gco:Integer>
    </mrc:bitsPerValue>
    <mrc:boundMin>
      <gco:Real>0.0000005</gco:Real> <! -- 0.5 微米 -->
    </mrc:boundMin>
    <mrc:boundMax>
      <gco:Real>0.00000075</gco:Real> <! -- 0.75 微米 -->
    </mrc:boundMax>
    <mrc:boundUnits>
      <gml:UnitDefinition>
        <gml:identifier codeSpace = "">m</gml:identifier>
      </gml:UnitDefinition>
    </mrc:boundUnits>
  </mrc:MI_Band>
</gie:bandinformation>
<gie:ratiometricCalibrationType>
  <gie:IE_RadiometricCalibrationType

```



```

codeList = "http://www.lmars.edu.cn/gie/codeList/gieCodeLists.xml # IE_RadiometricCalibrationType"
codeListValue = "AbsoluteRadiometricCalibration">AbsoluteRadiometricCalibration</gie:IE_RadiometricCalibrationType>
    </gie:ratiometricCalibrationType>
    <! --IE_OpticalImage-->
    <gie:opticalImageType>
        <gie:IE_OpticalImageType
codeList = "http://www.lmars.edu.cn/gie/codeList/gieCodeLists.xml # IE_OpticalImageType"
codeListValue = "PanchromaticImage">PanchromaticImage</gie:IE_OpticalImageType>
    </gie:opticalImageType>
    <gie:opticalSensorType>
        <gie:IE_OpticalSensorTypeCode
codeList = "http://www.lmars.edu.cn/gie/codeList/gieCodeLists.xml # IE_OpticalSensorTypeCode"
codeListValue = "Frame">Frame</gie:IE_OpticalSensorTypeCode>
    </gie:opticalSensorType>
</gie:IE_OpticalImage>

```

B.2 SAR 影像文件的标准内容模型 XML 描述实例

下面的 XML 内容是一个 SAR 影像文件的标准内容模型实例,描述了该 SAR 影像的空间参考、纠正模型、检查点、获取时间、波段和 SAR 传感器等信息。

```

<? xml version = "1.0" encoding = "UTF-8"?>
<gie:IE_SARAmplitudeData xsi:schemaLocation = "http://www.lmars.edu.cn/2017/gie
IE_SARData.xsd"
xmlns:gie = "http://www.lmars.edu.cn/2017/gie"
xmlns:gco = "https://schemas.isotc211.org/19103/-/gco/1.2"
xmlns:gcv = "http://www.isotc211.org/2005/gcv"
xmlns:msr = "https://schemas.isotc211.org/19115/-1/msr/1.3"
xmlns:smi = "https://schemas.isotc211.org/19130/-3/smi/1.1"
xmlns:mrc = "https://schemas.isotc211.org/19115/-1/mrc/1.3"
xmlns:gex = "https://schemas.isotc211.org/19115/-1/gex/1.3"
xmlns:mri = "https://schemas.isotc211.org/19115/-1/mri/1.3"
xmlns:mac = "https://schemas.isotc211.org/19115/-2/mac/2.2"
xmlns:mrl = "https://schemas.isotc211.org/19115/-1/mrl/1.3"
xmlns:gts = "http://www.isotc211.org/2005/gts"
xmlns:gml = "http://www.opengis.net/gml/3.2"
xmlns:epsg = "urn:x-ogp:spec:schema-xsd:EPSG:1.0:dataset"
xmlns:xsi = "http://www.w3.org/2001/XMLSchema-instance">
    <! -- CV_Coverage -->
    <gcv:commonPointRule>
        <gcv:CV_CommonPointRule
codeList = "http://www.lmars.edu.cn/gie/codeList/gieCodeLists.xml # CV_CommonPointRule"
codeListValue = "average">average</gcv:CV_CommonPointRule>

```

```

</gcv:commonPointRule>
<gcv:rangeType>
  <gco:RecordType>
    <gco:fieldType>
      <gco:FieldType>
        <gco:fieldName>
          <gco:CharacterString>intensity</gco:CharacterString>
        </gco:fieldName>
        <gco:fieldType>
          <gco:TypeName>
            <gco:aname>
              <gco:CharacterString>Integer</gco:CharacterString>
            </gco:aname>
          </gco:TypeName>
        </gco:fieldType>
      </gco:FieldType>
    </gco:fieldType>
  </gco:RecordType>
</gcv:rangeType>
<gcv:domainExtent>
  <gex:EX_Extent>
    <gex:geographicElement>
      <gex:EX_GeographicBoundingBox>
        <gex:westBoundLongitude>
          <gco:Decimal>115.5</gco:Decimal>
        </gex:westBoundLongitude>
        <gex:eastBoundLongitude>
          <gco:Decimal>115.56</gco:Decimal>
        </gex:eastBoundLongitude>
        <gex:southBoundLatitude>
          <gco:Decimal>32.2</gco:Decimal>
        </gex:southBoundLatitude>
        <gex:northBoundLatitude>
          <gco:Decimal>32.26</gco:Decimal>
        </gex:northBoundLatitude>
      </gex:EX_GeographicBoundingBox>
    </gex:geographicElement>
  </gex:EX_Extent>
</gcv:domainExtent>
<gcv:CRS><! -- 空间参考信息 -->
  <gml:GeodeticCRS gml:id = "epsg-crs-4490">
    <gml:identifier
gml:codeSpace = "OGP">urn:ogc:def:crs:EPSG::4490</gml:identifier>

```

```

    <gml:name>China Geodetic Coordinate System 2000</gml:name>
    <gml:remarks>Adopted July 2008. Replaces Xian 1980 (CRS code 4610).</gml:remarks>
    <gml:scope>Horizontal component of 3D system.</gml:scope>
    <gml:ellipsoidalCS xlink:href = "urn:ogc:def:cs:EPSG::6422" />
    <gml:geodeticDatum xlink:href = "urn:ogc:def:datum:EPSG::1043" />
  </gml:GeodeticCRS>
</gcv:CRS>
<! -- IE_ImageryandGriddedData -->
<gie:interpolationType>
  <gcv:CV_InterpolationType
codeList = "http://www.lmars.edu.cn/gie/codeList/gieCodeLists.xml # CV_InterpolationType"
codeListValue = "cubic">cubic</gcv:CV_InterpolationType>
  </gie:interpolationType>
  <gie:georectifiedSpatialInfo> <! -- 影像纠正信息 -->
    <gie:IE_Georectified>
      <gie:geolocationSource>
        <gie:IE_GeolocationSourceCode
codeList = "http://www.lmars.edu.cn/gie/codeList/gieCodeLists.xml # IE_GeolocationSourceCode"
codeListValue = "GCP">GCP</gie:IE_GeolocationSourceCode>
        </gie:geolocationSource>
        <gie:correctionModel>
          <gie:IE_GeometricCorrectionModelCode
codeList = "http://www.lmars.edu.cn/gie/codeList/gieCodeLists.xml # IE_GeometricCorrection-
ModelCode"
codeListValue = "CorrespondenceModel">CorrespondenceModel</gie:IE_GeometricCorrectionMod-
elCode>
        </gie:correctionModel>
        <gie:checkPoints> <! -- 检查点 -->
          <gie:IE_LocationGCP>
            <gie:locationGCP>
              <smi:SD_GCPLocation>
                <smi:locationGeographicPosition>
                  <gml:Point>
                    <gml:pos>115.53 32.23</gml:pos>
                  </gml:Point>
                </smi:locationGeographicPosition>
                <smi:locationGridPosition>
                  <gcv:CV_GridPoint>
                    <gcv:gridCoord>
                      <gcv:coordValues>3000 3000</gcv:coordValues>
                    </gcv:gridCoord>
                  </gcv:CV_GridPoint>
                </smi:locationGridPosition>
              </gie:locationGCP>
            </gie:checkPoints>
          </gie:IE_LocationGCP>
        </gie:georectifiedSpatialInfo>
      </gie:IE_Georectified>
    </gie:interpolationType>
  </gie:georectifiedSpatialInfo>
</IE_ImageryandGriddedData>

```

```

        </smi:SD_GCPLocation>
    </gie:locationGCP>
</gie:IE_LocationGCP>
</gie:checkPoints>
<! -- MD_Georectified_Type -->
<msr:numberOfDimensions>
    <gco:Integer>2</gco:Integer>
</msr:numberOfDimensions>
<msr:axisDimensionProperties>
    <msr:MD_Dimension>
        <msr:dimensionName>
            <msr:MD_DimensionNameTypeCode
codeList = "http://www.lmars.edu.cn/gie/codeList/gieCodeLists.xml#MD_DimensionNameTypeCode"
codeListValue = "column">column</msr:MD_DimensionNameTypeCode>
        </msr:dimensionName>
        <msr:dimensionSize>
            <gco:Integer>6000</gco:Integer>
        </msr:dimensionSize>
        <msr:resolution>
            <gco:Measure
gml:uom = "urn:ogc:def:uom:EPSG::9102">0.00001</gco:Measure>
        </msr:resolution>
    </msr:MD_Dimension>
    <msr:MD_Dimension>
        <msr:dimensionName>
            <msr:MD_DimensionNameTypeCode
codeList = "http://www.lmars.edu.cn/gie/codeList/gieCodeLists.xml#MD_DimensionNameTypeCode"
codeListValue = "row">row</msr:MD_DimensionNameTypeCode>
        </msr:dimensionName>
        <msr:dimensionSize>
            <gco:Integer>6000</gco:Integer>
        </msr:dimensionSize>
        <msr:resolution>
            <gco:Measure
gml:uom = "urn:ogc:def:uom:EPSG::9102">0.00001</gco:Measure>
        </msr:resolution>
    </msr:MD_Dimension>
</msr:axisDimensionProperties>
<msr:cellGeometry>
    <msr:MD_CellGeometryCode
codeList = "http://www.lmars.edu.cn/gie/codeList/gieCodeLists.xml#MD_CellGeometryCode"
codeListValue = "area">area</msr:MD_CellGeometryCode>
</msr:cellGeometry>

```

```

    <msr:transformationParameterAvailability>
      <gco:Boolean>false</gco:Boolean>
    </msr:transformationParameterAvailability>
    <msr:checkPointAvailability>
      <gco:Boolean>true</gco:Boolean>
    </msr:checkPointAvailability>
    <msr:cornerPoints>
      <gml:Point>
        <gml:pos>115.5, 32.26</gml:pos>
      </gml:Point>
    </msr:cornerPoints>
    <msr:cornerPoints>
      <gml:Point>
        <gml:pos>115.56, 32.2</gml:pos>
      </gml:Point>
    </msr:cornerPoints>
    <msr:pointInPixel>
      <msr:MD_PixelOrientationCode>centre</msr:MD_PixelOrientationCode>
    </msr:pointInPixel>
  </gie:IE_Georectified>
</gie:georectifiedSpatialInfo>
<! -- IE_Imagery -->
<gie:acquisitionTime>
  <gts:TM_Period>
    <gts:begin>
      <gts:TM_Position>
        <gts:date8601>2020-08-01</gts:date8601>
      </gts:TM_Position>
    </gts:begin>
    <gts:end>
      <gts:TM_Position>
        <gts:date8601>2020-08-01</gts:date8601>
      </gts:TM_Position>
    </gts:end>
  </gts:TM_Period>
</gie:acquisitionTime>
<gie:isMosaic>
  <gco:Boolean>false</gco:Boolean>
</gie:isMosaic>
<gie:numberofBands>
  <gco:Integer>1</gco:Integer>
</gie:numberofBands>
<! -- 波段描述 -->

```

```

    <gie:bandinformation>
      <mrc:MI_Band>
        <mrc:bitsPerValue>
          <gco:Integer>16</gco:Integer>
        </mrc:bitsPerValue>
        <mrc:boundMin>
          <gco:Real>0.035</gco:Real> <! -- 7 厘米 -->
        </mrc:boundMin>
        <mrc:boundMax>
          <gco:Real>0.07</gco:Real> <! -- 3.5 厘米 -->
        </mrc:boundMax>
        <mrc:boundUnits>
          <gml:UnitDefinition>
            <gml:identifier codeSpace = "">m</gml:identifier>
          </gml:UnitDefinition>
        </mrc:boundUnits>
      </mrc:MI_Band>
    </gie:bandinformation>
    <gie:ratiometricCalibrationType>
      <gie:IE_RadiometricCalibrationType
codeList = "http://www.lmars.edu.cn/gie/codeList/gieCodeLists.xml # IE_RadiometricCalibra-
tionType"
codeListValue = "AbsoluteRadiometricCalibration">AbsoluteRadiometricCalibration</gie: IE_
RadiometricCalibrationType>
    </gie:ratiometricCalibrationType>
    <! -- IE_SARData -->
    <gie:farRangeIncidenceAngle>
      <gco:Real>25</gco:Real>
    </gie:farRangeIncidenceAngle>
    <gie:nearRangeIncidenceAngle>
      <gco:Real>25</gco:Real>
    </gie:nearRangeIncidenceAngle>
    <gie:IE_SARImagingModeCode
codeList = "http://www.lmars.edu.cn/gie/codeList/gieCodeLists.xml # IE_SARImagingModeCode"
codeListValue = "SpotlightMode">SpotlightMode</gie: IE_SARImagingModeCode>
    <gie:sarModeDescription>
      <gco:CharacterString>滑块聚束(SL)</gco:CharacterString>
    </gie:sarModeDescription>
    <gie:waveLength>
      <gco:Real>0.06</gco:Real>
    </gie:waveLength>
    <! -- IE_SARAmplitudeData -->
    <gie:looksInAzimuth>

```

```

    <gco:Integer>1</gco:Integer>
  </gie:looksInAzimuth>
  <gie:looksInRange>
    <gco:Integer>1</gco:Integer>
  </gie:looksInRange>
  <gie:IE_SARSensor>
    <gie:azimuthBandwidth>
      <gco:Real>741.833712</gco:Real>
    </gie:azimuthBandwidth>
    <gie:dopplerCentroid>
      <gie:IE_SARDopplerCentroidParameter>
        <gie:polynomial_coefficient_0>
          <gco:Real>3.642657</gco:Real>
        </gie:polynomial_coefficient_0>
        <gie:polynomial_coefficient_1>
          <gco:Real>0.004208</gco:Real>
        </gie:polynomial_coefficient_1>
        <gie:polynomial_coefficient_2>
          <gco:Real>-0.000004</gco:Real>
        </gie:polynomial_coefficient_2>
        <gie:reference_point>
          <gco:Real>2993.248253</gco:Real>
        </gie:reference_point>
        <gie:time>
          <gco:DateTime>2022-2-18T11:29:21.999999</gco:DateTime>
        </gie:time>
      </gie:IE_SARDopplerCentroidParameter>
    </gie:dopplerCentroid>
    <gie:oneWayTimeToFirstRangePixel>
      <gco:Real>11.12</gco:Real>
    </gie:oneWayTimeToFirstRangePixel>
    <gie:pulseRepetitionFrequency>
      <gco:Real>45.000000</gco:Real>
    </gie:pulseRepetitionFrequency>
    <gie:rangeBandwidth>
      <gco:Real>60.0</gco:Real>
    </gie:rangeBandwidth>
    <gie:rangeSamplingRate>
      <gco:Real>66.666667</gco:Real>
    </gie:rangeSamplingRate>
    <gie:timeOfFirstAzimuthLine>
      <gco:DateTime>2018-10-16T10:26:25.000000</gco:DateTime>
    </gie:timeOfFirstAzimuthLine>

```

```

    <gie:orbitParameter>
      <gie:velocityX>
        <gco:Real>2770.806531</gco:Real>
      </gie:velocityX>
      <gie:velocityY>
        <gco:Real>-2873.156907</gco:Real>
      </gie:velocityY>
      <gie:velocityZ>
        <gco:Real>6431.850361</gco:Real>
      </gie:velocityZ>
      <gie:x>
        <gco:Real>-1987636.767214</gco:Real>
      </gie:x>
      <gie:y>
        <gco:Real>5894741.209148</gco:Real>
      </gie:y>
      <gie:z>
        <gco:Real>3478175.935571</gco:Real>
      </gie:z>
      <gie:time>
        <gco:DateTime>2022-2-18T11:28:10.441457</gco:DateTime>
      </gie:time>
    </gie:orbitParameter>
  </gie:IE_SARSensor>
</gie:IE_SARAmplitudeData>

```


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